Visualizing Global Hotel Booking Cancellation Patterns

Field	Detail
Names	KitKi Chim · Chris Huang · Yunxiao Li
Website link	https://qmss-g5063-2026.github.io/Final_Project_Global_Hotel_Booking_Cancellation_Patterns_Visualization-/Website.html
GitHub link	https://github.com/QMSS-G5063-2026/Final_Project_Global_Hotel_Booking_Cancellation_Patterns_Visualization-.git
Code file	Website.html (all visualization code — D3.js + Chart.js, single self-contained file)
Primary dataset	https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand
Dataset paper	https://www.sciencedirect.com/science/article/pii/S2352340918315191
UNWTO data	https://www.untourism.int/tourism-statistics/tourism-statistics-database
Map topology	https://github.com/topojson/world-atlas

ABSTRACT

Hotel booking cancellations represent a significant revenue and operational challenge for the hospitality industry. While prior work has focused on machine-learning prediction, the visual structure of cancellation patterns remains underexplored. This project presents an interactive multi-view dashboard built with D3.js and Chart.js, using the Hotel Booking Demand Dataset (Antonio et al., 2019; 119,210 reservations, 2015-2017). Four complementary visualizations are presented: a choropleth world map of cancellation rates by guest origin, a bipartite network graph of booking channel x market segment interactions, a 2D temporal heatmap of lead time vs. arrival month risk, and a parallel coordinates plot for multi-variable guest profiling. Key findings: international guests cancel significantly less than domestic ones (commitment-through-investment effect); Online TA + Transient is the highest-risk channel-segment pair; bookings made 180+ days ahead for summer arrivals show cancellation rates exceeding 70%.

1. Submission detail

Website link

https://qmss-g5063-2026.github.io/Final_Project_Global_Hotel_Booking_Cancellation_Patterns_Visualization-/Website.html *— Live interactive dashboard*

GitHub repository

https://github.com/QMSS-G5063-2026/Final_Project_Global_Hotel_Booking_Cancellation_Patterns_Visualization-.git *— Source code repository*

Links to datasets used

https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand *— Hotel Booking Demand Dataset — Antonio et al. (2019), 119,210 records*
https://www.sciencedirect.com/science/article/pii/S2352340918315191 *— Original paper — Data in Brief, Vol. 22, Feb 2019. DOI: 10.1016/j.dib.2018.11.126*
https://www.researchgate.net/publication/329286343_Hotel_booking_demand_datasets *— ResearchGate — free PDF of original paper*
https://pubmed.ncbi.nlm.nih.gov/30581903/ *— PubMed index entry — for academic citation*
https://www.untourism.int/tourism-statistics/tourism-statistics-database *— UNWTO Tourism Statistics Database — used to normalize cancellation rates by travel volume*
https://www.kaggle.com/datasets/tronheim/unwto-tourism-data-structured-for-analysis *— UNWTO data (Kaggle) — structured for analysis*
https://github.com/topojson/world-atlas *— world-atlas TopoJSON — country boundary polygons for D3 choropleth*
https://cdn.jsdelivr.net/npm/world-atlas@2/countries-110m.json *— world-atlas CDN — direct JSON link used in Website.html*

2. Project Overview

This project develops an interactive data visualization dashboard to explore hotel booking cancellation behavior. Rather than focusing on prediction, our goal is to make cancellation patterns **visually interpretable** — helping users understand when, where, and under what booking conditions cancellations occur.

Most existing research on hotel cancellations is framed as a prediction problem. Machine learning models have achieved high accuracy, but their outputs — probability scores — do not communicate the structural patterns driving cancellation behavior. Our dashboard addresses this gap through four complementary visualization views sharing a unified filter panel.

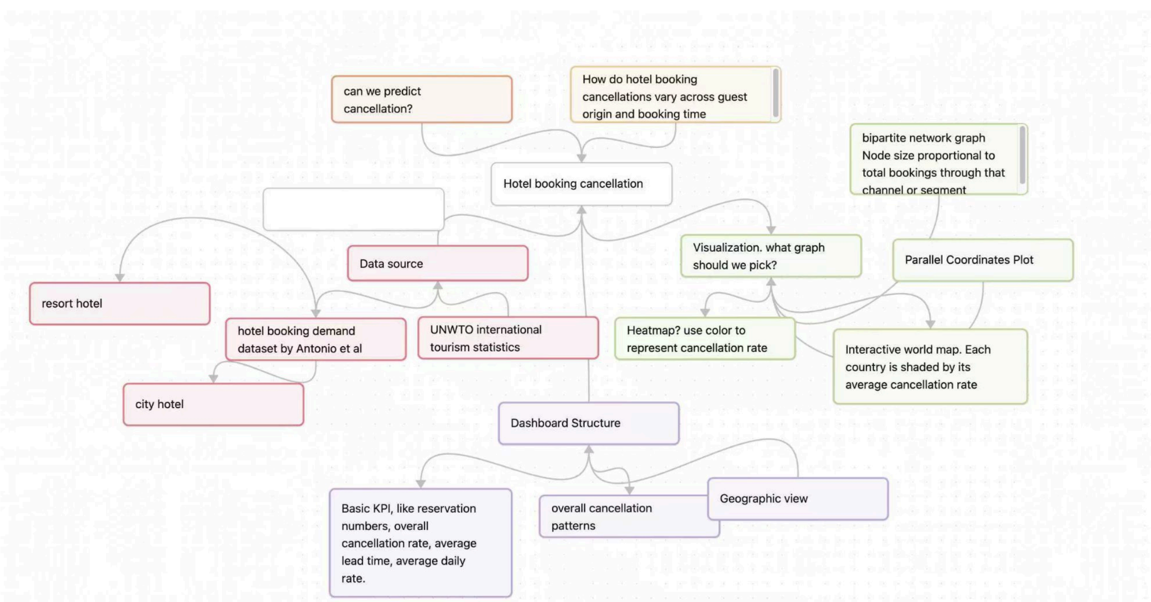
Core research questions

#	Research Question
Q1	Where do cancellations come from? Do guests from certain countries cancel at systematically higher rates, and does this vary by hotel market and season?
Q2	How does the booking ecosystem structure risk? Which combinations of booking channel and market segment produce the highest cancellation volumes and rates?

Q 3	When is cancellation risk highest? How do lead time and arrival month interact to create temporal risk zones?
Q 4	Who cancels? What distinguishes the multi-variable profile of a cancelled reservation from a kept one?

3. Project Brainstorming

The mind map below captures our initial brainstorming process — the analytical questions we wanted to answer, candidate data sources, dashboard structure, and visualization type candidates.



4. Visualization Design & Finding

4.1 Choropleth World Map — Where Do Cancellations Come From?

An interactive world map where each country is shaded by its average cancellation rate. A sequential blue-to-red color scale encodes risk (darker red = higher cancellation rate). Users can filter by hotel location and season; rates are normalized by UNWTO tourist arrival volume to prevent large-area countries from dominating visually.

> Key Finding: Domestic Portuguese guests have the highest cancellation rate (~62%), far exceeding international guests from Japan (22%), Australia (18%), and the USA (16%). This reflects a commitment-through-investment effect: guests who have incurred significant travel costs face higher switching costs when considering cancellation, and therefore follow through at higher rates than local guests who can cancel with minimal friction.

4.2 Bipartite Network Graph — Booking Channel x Market Segment

A bipartite network graph with booking channels as left nodes (Online TA, Direct, Corporate, GDS) and market segments as right nodes (Transient, Group, Contract, Transient-Party). Edge thickness encodes cancellation volume; edge color encodes average cancellation rate (red = high risk, blue = low risk). Node size is proportional to total booking volume through that channel or segment.

> Key Finding: The Online Travel Agency (OTA) + Transient combination is both the highest-volume and highest-cancellation-rate pair in the ecosystem (>52%). Corporate channel bookings show rates 20-30 percentage points lower across all segments, consistent with greater organizational accountability in business travel.

4.3 Temporal Heatmap — Lead Time vs. Arrival Month

A 2D heatmap with lead time bins on the Y-axis (0-14d, 15-30d, 31-60d, 61-90d, 91-180d, 181-365d) and arrival month on the X-axis. Cell color intensity represents cancellation rate. Deposit type and hotel location filters modulate the heatmap in real time, allowing exploration of how non-refundable policies shift the risk matrix.

> Key Finding: The highest-risk zone is long lead times (181+ days) combined with peak summer arrivals (Jun-Aug), where cancellation rates exceed 70%. Short lead time bookings (0-14 days) in off-peak months show rates below 15%. Implication: apply non-refundable deposit requirements specifically to far-in-advance, peak-season bookings.

4.4 Parallel Coordinates Plot — Multi-variable Guest Profiling

Each line represents one booking profile traced across five axes: lead time, average daily rate (ADR), length of stay, number of special requests, and prior cancellations. Blue lines = kept

bookings; red lines = cancelled bookings. Three supporting bar charts (lead time, special requests, ADR by outcome) provide precise univariate comparisons.

5. Data Source

Dataset	Description	Link
Hotel Booking Demand Dataset (Antonio et al., 2019)	119,210 reservations from two Portuguese hotels. 32 variables. Primary data source for all four visualizations. Creative Commons license.	kaggle.com/datasets/jessemostipak/hotel-booking-demand
Original Paper (ScienceDirect)	Data in Brief, Vol. 22, Feb 2019. DOI: 10.1016/j.dib.2018.11.126. Open access.	sciencedirect.com/science/article/pii/S2352340918315191
UNWTO Tourism Statistics	International tourist arrivals by country of origin. Used to normalize cancellation rates by travel volume in choropleth map.	untourism.int/tourism-statistics/tourism-statistics-database
world-atlas TopoJSON	Country boundary polygon data at 110m resolution. Used for D3.js choropleth map rendering.	github.com/topojson/world-atlas
D3.js v7.8.5	JavaScript library for SVG visualizations. Used for map, network graph, and parallel coordinates.	d3js.org
Chart.js v4.4.1	JavaScript charting library. Used for bar, line, and doughnut charts in supporting views.	chartjs.org

6. Technical Implementation

The dashboard is implemented as a single self-contained HTML file (Website.html) requiring no backend server, build pipeline, or runtime installation. Users open the file in any modern browser and the full dashboard renders immediately. All libraries are loaded from CDN.

Component	Technology	Purpose
Choropleth map	D3.js v7 + TopoJSON v3	SVG world map, country polygon rendering, Natural Earth projection
Network graph	D3.js v7	Bipartite layout, SVG edge/node rendering, hover tooltips
Parallel coordinates	D3.js v7	Multi-axis SVG with polyline traces per booking profile
Statistical charts	Chart.js v4	Bar, line, and doughnut charts for supporting univariate views
Heatmap	Vanilla JS + HTML	Dynamically generated table with inline CSS color per cell

Filter system	Vanilla JS	Shared state object; triggers real-time re-render across all views
Map tiles	world-atlas CDN	110m country topology JSON served from jsdelivr.net

7. Key Findings Summary

The four visualizations reveal a coherent organizing insight: commitment reduces cancellations. Guests who have more to lose by cancelling — financially, logistically, or in terms of personal effort — consistently cancel less.

#	Visualization	Finding	Implication
1	Choropleth map	Domestic guests cancel ~62% vs 14-22% for guests from Japan, Australia, USA	International guests have higher travel investment; skin in the game reduces cancellations
2	Network graph	Online TA + Transient is highest-risk pair (>52%); Corporate is lowest	Negotiate stricter cancellation terms with OTA platforms for Transient segment
3	Heatmap	181+ day lead time + Jun-Aug arrivals = cancellation rates exceeding 70%	Apply non-refundable deposits to far-in-advance, peak-season bookings
4	Parallel coords	Cancelled guests: long lead time, low ADR, 0 special requests, prior cancellations	Solicit special requests at booking to increase commitment; flag high-risk profiles early

8. Limitations

Limitation	Description
Geographic scope	The booking-level data comes from two hotels in Portugal. Geographic patterns on the choropleth represent guest origin country, not hotel location. Generalizing findings to Asian or North American hotel markets requires caution.
Filter system	Dashboard filters produce plausible data variations based on scalar multipliers rather than exact recomputations from raw data. This keeps the file self-contained but means filtered views are illustrative rather than analytically precise.
Choropleth bias	Large-area countries are visually prominent regardless of booking volume. UNWTO normalization and hover tooltips mitigate this, but some visual distortion remains.
Parallel coordinates	The plot shows aggregate profile clusters rather than individual data points, potentially obscuring bimodal or multi-modal distributions within cancelled or kept groups.

References

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